NOT MEASUREMENT SENSITIVE

A-A-33B
28 July 2009
SUPERSEDING
A-A-33A
17 October 1997

COMMERCIAL ITEM DESCRIPTION

SCALE REMOVING COMPOUND (LIQUID)

The General Services Administration has authorized the use of this commercial item description, for all federal agencies.

1. SCOPE. This CID covers a liquid form scale removing compound, hereafter referred to as "compound", intended for removing scale deposits from environmental surfaces, commercial dishwashing machines, and boiler coilers. This compound can be used in laboratories, food handling and processing areas, veterinary areas, and dairy plants.

2. SALIENT CHARACTERISTICS.

2.1 Materials. Unless otherwise specified herein, materials shall be in accordance with the manufacturer's material specifications. The compound shall be a uniform liquid, free of sediments and suspended particles, and shall contain a corrosion inhibiting agent and a wetting agent.

2.1.1 Phosphoric acid (as P_2O_5). The compound shall have a phosphoric acid content of not less than 30.0% and not more than 40.0% when determined in accordance with ASTM D820 Electrometric Titration Method.

2.1.2 Hydrogen-ion concentration (pH). The compound shall have a pH of not more than 2.0 when determined in accordance with ASTM E70.

2.1.3 Citric acid. The compound shall have a citric acid content of more than 6% when determined in accordance with ASTM D4608.

2.1.4 Hydrochloric acid (HCl). The compound shall contain not more than 10% HCl when determined in accordance with the manufacturer's standard practice.

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any data that may improve this document should be sent to: Defense Supply Center Richmond, ATTN: DSCR-VEB, 8000 Jefferson Davis Highway, Richmond, VA 23297-5616 or by e-mail to STDZNMGT@DLA.MIL. Since contact information can change, you may verify the currency of this address information using the ASSIST database at http://assist.daps.dla.mil.

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2.1.5 Excluded ingredients. The compound shall not contain sulfuric acid or organic solvents.

2.2 Surface tension. The compound shall have a surface tension not greater than 8.08 x 10^{-3} poundals per inch (44.0 dynes per centimeter) when measured in accordance with ASTM D1331 using a sample at 77°F (25°C) in a 1% aqueous solution.

2.3 Corrosive effects on steel. The compound, in a 1% distilled aqueous solution, shall have a corrosive effect on steel of not more than 0.15% loss of mass on 2 inch by 2 inch by 0.053 inch thick (51mm X 51mm X 0.89mm thick) coupons of hot rolled steel after a 24 hour period of immersion in the compound.

2.4 Nonvolatile matter. The compound shall have a nonvolatile matter content of not less than 30.0% at 221°F (105°C) when determined in accordance with ASTM D2834.

2.5 Foaming. The compound shall exhibit no foam at temperatures above 100°F (38°C).

2.6 Identification and markings. Identification and markings shall be permanent and legible and shall include, as a minimum, the manufacturer's identification code (CAGE) and labeling with instructions for use.

3. REGULATORY REQUIREMENTS.

3.1 Hazardous materials. The compound shall meet all applicable Federal codes and regulations concerning hazardous materials, including package marking.

4. PRODUCT CONFORMANCE PROVISIONS.

4.1 Product conformance. The product shall meet the salient characteristics of this CID, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial market. The Government reserves the right to require proof of such conformance.

4.2 Market acceptability. The compound shall have been sold previously to the Government or to the commercial market in order to qualify for acceptability under this CID

5. PACKAGING. Preservation, packaging, and marking shall be as specified in the contract or order (see 6.2).

6. NOTES.

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

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6.1 Referenced documents.

American Society for Testing and Materials (ASTM)

D820 –	Standard Test Methods for Chemical Analysis of Soaps Containing Synthetic Detergents
D1331 –	Standard Test Methods for Surface and Interfacial Tension of Solutions of Surface-Active Agents
D2834 –	Standard Test Method for Nonvolatile Matter (Total Solids) in Water-Emulsion Floor Polishes, Solvent-Based Floor Polishes, and Polymer Emulsion Floor Polishes
D4608 –	Standard Test Method for Citrates in Detergents
E70 –	Standard Test Method for pH of Aqueous Solutions with Glass Electrode

Copies of these documents can be obtained online at http://www.astm.org or from American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.

6.2 Ordering data. Acquisition documents must specify the following:

a. Title, number, and date of this CID.
b. The specific issue of individual documents referenced, if required.
c. Packing requirements.

6.3 Cross-reference data. Scale removing compound conforming to this CID is interchangeable/substitutable with scale removing compound conforming to P-S-120B, type II, dated April 8, 1977 (cancelled October 17, 1997 with supersession by this document).

MILITARY INTERESTS:

Custodian:
DLA – GS

CIVIL AGENCY COORDINATING ACTIVITY:
GSA-FAS

Preparing Activity:
DLA – GS3

(Project 6850-2009-009)

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